ALAMEIN INTERNATIONAL UNIVERSITY

Faculty of Computer Science & Engineering

GP2 Mid-Term Defense — Spring 2026

SecureSight

Smart Surveillance System to Detect Unusual Activities in Real Time

Ahmed Mostafa | Mohamed Hassona | Abdullah Abdalgawad | Omnia Adel | Mariam Elrafei | Khalid Abdelsalam | Tasneem Ibrahim

Supervised by Dr. Essam Abdellatef

ResNet-50 · R(2+1)D · Vision Transformer · UCF-Crime Dataset · Real-Time Anomaly Detection

SECTION 1 · INTRODUCTION

Project Overview

Goal

Transform passive CCTV networks into active, real-time anomaly detection systems capable of immediate alert routing.

Core Technology

Dual-stream deep learning: ResNet-50 (spatial) + R(2+1)D (temporal) fused with a Vision Transformer sequence encoder.

Dataset

UCF-Crime benchmark — 14 crime classes, 128 hours, trained on precise clip-level temporal annotations.

Output

Multi-class anomaly classification with role-differentiated alert routing (Police, EMS, Fire) via MQTT protocol.

Peak Performance: 91.6% Accuracy | 91.5% F1 | 9.86ms Latency | Mean AUC 0.89

SECTION 1 · INTRODUCTION

Problem Statement

01

Passive Utilization

Over 1 billion CCTV cameras globally capture footage that is never analyzed. Human operators miss events due to cognitive fatigue within 30 minutes of monitoring.

02

Modality Silos

Existing systems rely on spatial OR temporal features independently — failing to jointly characterize what anomalies look like AND how they move.

03

Deployment Gap

Academic models optimize metrics on static test sets, ignoring production requirements: alert routing, fault tolerance, privacy compliance, and low latency.

SECTION 1 · INTRODUCTION

Motivation

1B+
CCTV cameras globally by 2025

30min
Until operator attention degrades

72%
Citizens feel unsafe in crowded public areas

0%
Footage reviewed in real time

The Paradox

Cities possess unprecedented visual capacity yet lack simultaneous processing capability. SecureSight resolves this by synthesizing CNN spatial precision with ViT long-range temporal reasoning into a unified, low-latency proactive surveillance agent.

Scale Overload · Delayed Reactive Response · Cognitive Fatigue → SecureSight addresses all three

SECTION 1 · INTRODUCTION

Project Objectives

1 **Dual-Stream Neural Pipeline**

Architect a parallel feature extraction framework using pre-trained ResNet-50 (spatial) and R(2+1)D (temporal) backbones.

2 **Lightweight Modality Fusion**

Implement an optimized projection & normalization pipeline concatenating 2048-D + 512-D embeddings into a 512-D joint token.

3 **Long-Range Attention Engine**

Integrate a compact 4-block Vision Transformer (ViT) encoder with multi-head self-attention across an 8-clip sliding window.

4 **Precise Temporal Supervision**

Eliminate noisy video-wide weak labels by using clip-level temporal annotations from the UCF-Crime benchmark.

5 **False-Alarm Safeguards**

Implement 60% confidence threshold + 5-second temporal consistency check to suppress false positives with minimal latency.

SECTION 1 · INTRODUCTION

Project Contributions

Novel Hybrid Architecture

First application of ResNet-50 + R(2+1)D dual-stream fusion with a compact 4-block ViT encoder for crime video anomaly detection on UCF-Crime.

Clip-Level Supervision

Restructured UCF-Crime training around precise temporal annotations, eliminating weak-label noise and achieving significant accuracy improvements over baselines.

Production Engineering

Embedded the inference pipeline into an Express/Flask decoupled web architecture with MQTT alert routing, offline edge buffering, and ephemeral privacy processing.

9-Class Merged Taxonomy

Compressed 14 sparse UCF-Crime classes into 9 semantically coherent categories, reducing the classification head and improving generalization.

Statistical Rigor

Conducted 25 evaluation runs across 5 hyperparameter batches providing confidence intervals, stability rankings, and deployment recommendations.

SECTION 1 · INTRODUCTION

Challenges Addressed

⚠ **Environmental Variance**

→ Robust preprocessing with normalization, multi-scale augmentation, and handling of day/night/weather variations in surveillance footage.

⚠ **Dual-Modality Representation**

→ Parallel ResNet-50 + R(2+1)D factorized architecture captures orthogonal appearance and motion streams without multiplicative pipeline cost.

⚠ **Severe Class Imbalance**

→ Normal class (948k frames) vs sparse crimes (7k–40k frames). Mitigated via stratified sampling, inverse-weight loss, and redundant normal video removal.

⚠ **Label Noise (Weak Supervision)**

→ Moved to precise clip-level temporal annotations from the official UCF-Crime GitHub, eliminating 10-min bag-level noise present in MIL baselines.

⚠ **Latency-Accuracy Trade-Off**

→ R(2+1)D factorization reduces 3D conv cost. Merged 9-class taxonomy compresses output head. Achieves 42.45ms latency at 91.6% accuracy.

SECTION 2 · DATASET

Dataset Overview — UCF-Crime

14	9	128h	1,900
Original Classes	Merged Classes	Total Video	Surveillance Videos

Original 14 Classes:

Abuse
Arrest
Arson
Assault
Burglary
Explosion
Fighting
Normal
RoadAccidents
Robbery
Shooting
ShopLifting
Stealing
Vandalism

Merged 9-Class Taxonomy:

Violence (Assault + Fighting + Abuse)

Theft (ShopLifting + Stealing + Burglary + Robbery)

Arson (Arson)

Explosion (Explosion)

Arrest (Arrest)

RoadAccidents (RoadAccidents)

Shooting (Shooting)

Vandalism (Vandalism)

Normal (Normal)

SECTION 2 · DATASET

Data Collection Pipeline

1

Raw UCF-Crime

128h surveillance video
14 crime categories
+ Normal activity

2

Official Temporal Annotations

Clip-level start/end timestamps from Xuange923 GitHub

3

Frame Extraction

5 FPS sampling
16-frame clip volumes
112×112 resize

4

Class Merging

14 → 9 merged taxonomy based on visual similarity

5

Stratified Split

70% Train
15% Validation
15% Test

SECTION 2 · DATASET

Dataset Challenges

Extreme Class Imbalance

Normal class: 948k training frames vs Shooting: 7.1k. A 133:1 ratio forces gradient bias toward normality, causing false negatives on critical anomalies.

Weak Label Noise (MIL)

Traditional bag-level labels annotate a full 10-min video as anomalous. This injects massive normal-frame noise into anomaly gradients during backpropagation.

Environmental Variance

Real surveillance footage has fixed camera angles, variable frame rates, night/day lighting shifts, weather occlusions, and compressed low-resolution streams.

Visual Class Ambiguity

Arrest and Robbery share similar visual patterns with Normal street activity. Low AUC (0.78 Arrest, 0.82 Robbery) confirms the difficulty of discrimination in low-res CCTV.

SECTION 2 · DATASET

Data Preprocessing Pipeline

1

Raw Video

UCF-Crime surveillance footage with temporal annotation timestamps

2

Frame Cleaning

Remove corrupted frames, normalize aspect ratios, handle variable FRs

3

Enhancement

Resize to 224×224 (spatial) & 112×112 (temporal), ImageNet normalization

4

Augmentation

Temporal jitter, horizontal flip, random crop, brightness variation

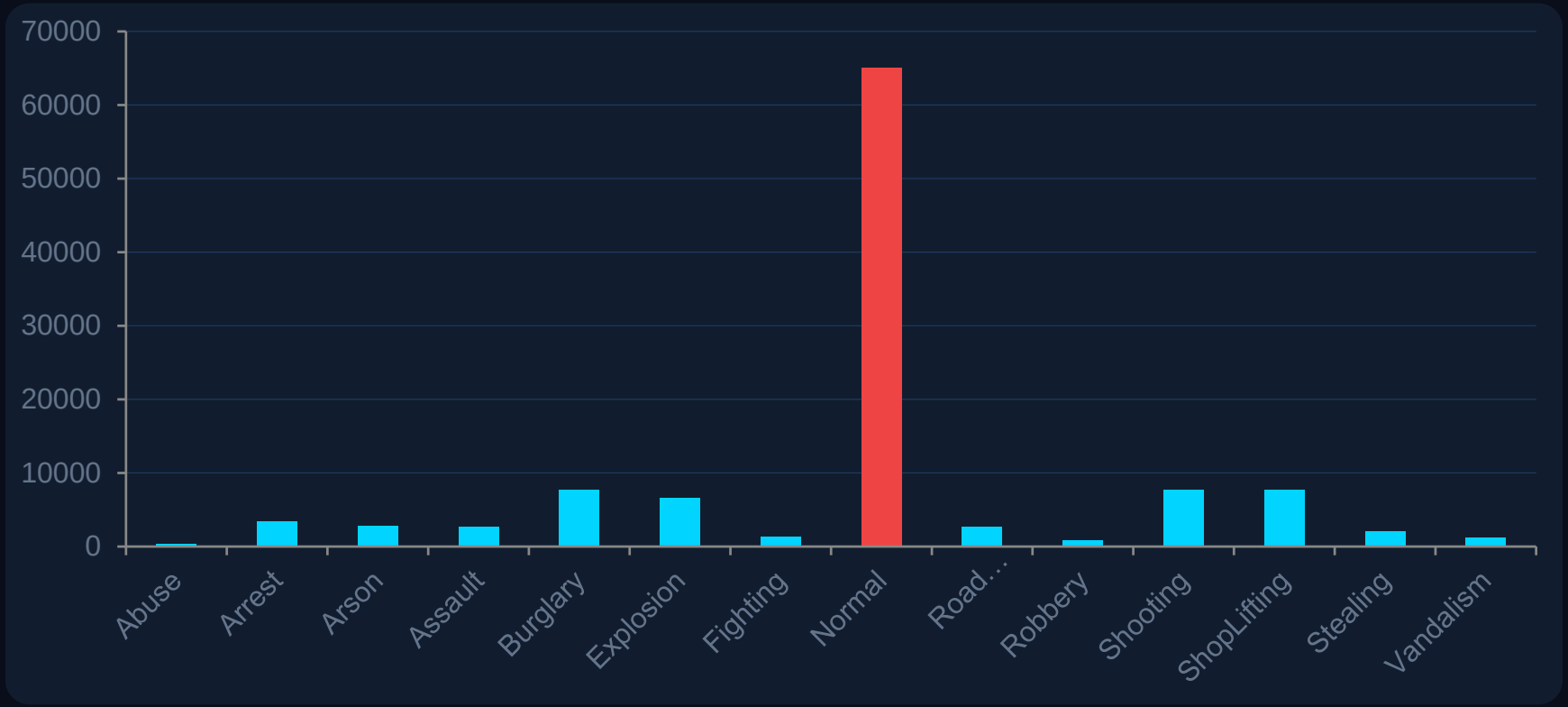
5

Final Split

Stratified 70/15/15 train/val/test with class balance verification

SECTION 2 · DATASET

Dataset Statistics — Training Distribution



↑ Normal class (65k frames highlighted in red) dominates — 133× more frames than Shooting class

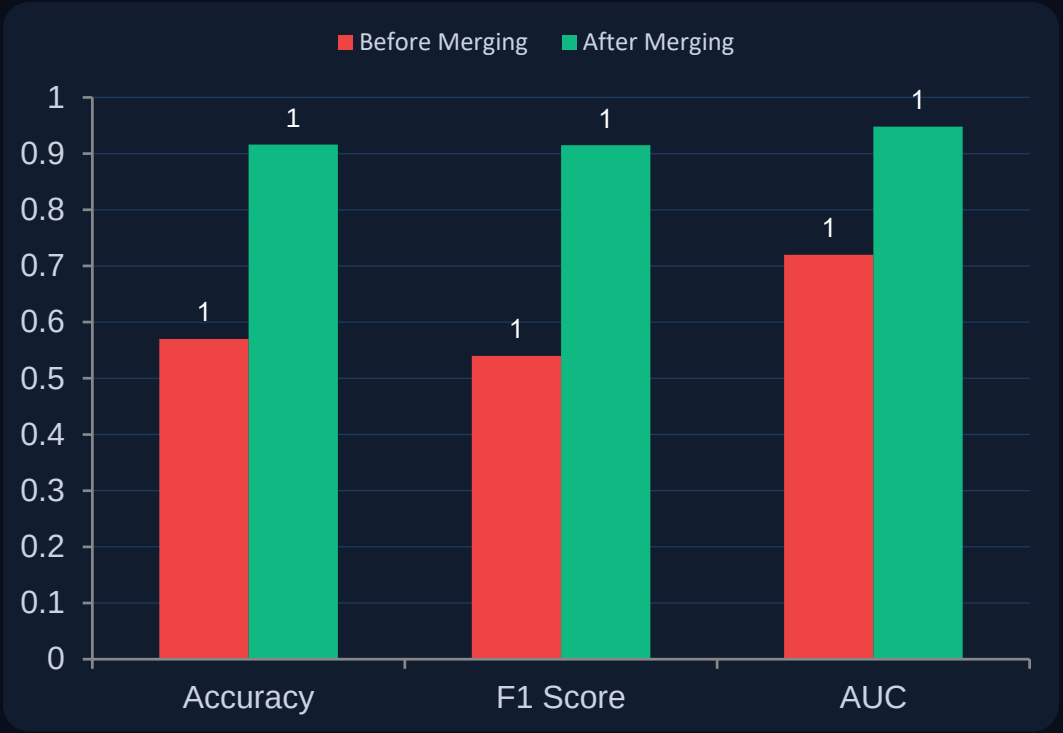
SECTION 2 · DATASET

Before vs. After Dataset Merging & Optimization

	BEFORE	AFTER
Classes:	14 sparse categories	9 merged taxonomy
Normal Frames:	948,000 (training)	Downsampled (balanced)
Shooting Frames:	7,100 (training)	7,100 (preserved)
Label Quality:	Video-level weak (MIL)	Clip-level temporal annot.
Class Balance:	Severe 133:1 imbalance	Inverse-weight sampling
Accuracy Baseline:	50–60% (weak sup.)	**Accuracy Achieved:** 74.3% → 91.6% (best run)
Split Method:	Ad-hoc partitioning	Stratified 70/15/15

SECTION 2 · DATASET

Impact of Dataset Merging on Model Performance



Gradient Balance

Inverse-weight sampling ensures backprop penalizes errors on rare crime classes 5–133× more than Normal, eliminating normality bias.

Semantic Clarity

Merging 14 → 9 classes removed intra-class ambiguity (Robbery vs Stealing) enabling the model to learn sharper decision boundaries.

Convergence Speed

Cleaner label structure combined with clip-level supervision accelerated convergence from 60 epoch budgets to stable performance by epoch 30–35.

SECTION 3 · SYSTEM WORKFLOW

Complete System Workflow

1 **Input CCTV** — 2 **Frame Extraction** — 3 **Dual-Branch Feature Extract** — 4 **ViT Sequence Modeling** — 5 **Threat Analysis** — 6 **Alert Routing** — 7 **Dashboard Output**

Safety Gate 1: 60% Confidence Threshold **Safety Gate 2:** 5-Second Temporal Stability Filter

Police

Violence, Shooting,
Burglary, Robbery, Assault

EMS/Hospital

Road Accidents,
Abuse, Explosion

Fire Dept.

Arson, Explosion,
Hazardous Events

SECTION 3 · SYSTEM WORKFLOW

Detailed Workflow Breakdown

1. Ingestion

RTSP stream via multi-threaded Python pipeline. Frames downsampled to model processing frequency. GPU VRAM loaded for zero-overhead inference.

2. Preprocessing

Spatial: 224×224×3 (ImageNet norm). Temporal: 112×112×3×16 frames at 5 FPS. CPU handles resize/normalize while GPU processes previous batch.

3. Dual Extraction

ResNet-50 → 2048-D spatial vector. R(2+1)D factorized → 512-D temporal vector. Concatenated → 2560-D joint embedding → Projected to 512-D via Wp.

4. ViT Encoding

8 consecutive clip embeddings form 8×512 sliding matrix. [CLS] token prepended. 4 transformer blocks with MHSA + FFN. Final z_4^o → classification head.

5. Decision Engine

Confidence ≥60% required. If met, 5-sec temporal stability tracker activated. Only sustained anomalies trigger alert — transient noise filtered out.

6. Alert Hub

MQTT protocol routes JSON payloads to role-specific channels. Database writes on confirmed event. Frontend React dashboard updated via long-polling.

SECTION 3 · SYSTEM WORKFLOW

System Architecture Overview

Layer 1 · Data Capture

- Multi-threaded RTSP ingestion
- Local IP camera network
- Edge frame buffer

Layer 2 · Deep Learning Engine

- Flask + PyTorch inference server
- ResNet-50 + R(2+1)D dual-branch
- 4-block ViT encoder

Layer 3 · Backend Gateway

- Express / Node.js API
- MySQL relational storage
- 1s TTL memory cache

Layer 4 · Client Interface

- React + Vite frontend
- Real-time long-polling loop
- Color-coded alert dashboard

Languages: Python · JavaScript (Node.js / React) | AI: PyTorch · torchvision | DB: MySQL | Hardware: NVIDIA RTX 4070

SECTION 3 · SYSTEM WORKFLOW

End-to-End Processing Pipeline

Total: 42.45ms per spatiotemporal window · Throughput: 6.76 windows/sec



SECTION 4 · MODEL SELECTION

Candidate Models Evaluated

Model	Type	Strengths	Weaknesses	Selected?
ResNet-50 + R(2+1)D + ViT	Dual-stream Hybrid	High accuracy, long-range temporal, modular	Higher latency than EfficientNet	✔ YES
EfficientNet-B0	Efficient CNN	Fast, lightweight, good accuracy/param ratio	Spatial only — no temporal modeling	✗ Trial
3D CNN (I3D)	Spatiotemporal	Joint space-time modeling	Massive parameter count, slow inference	✗ No
LSTM + CNN	Recurrent	Temporal memory, proven for sequences	Sequential — no parallel attention, slow	✗ No
MIL Baseline (Sultani)	Weak supervised	Standard benchmark, simple training	Video-level labels, high false negatives	✗ Baseline
GWO / PSO Optimizers	Evolutionary	Global optimization, no gradient needed	Did not outperform ViT baseline	✗ No

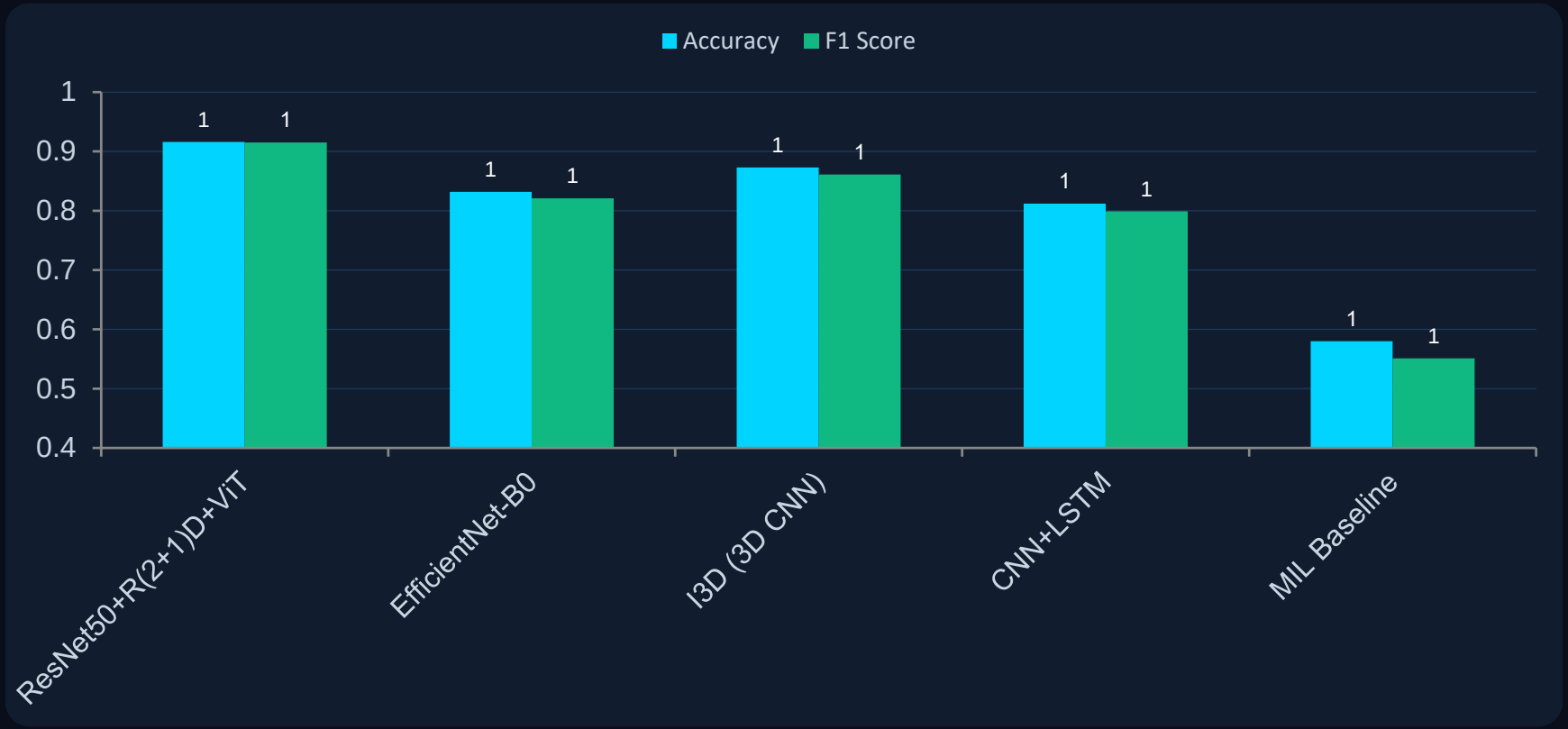
SECTION 4 · MODEL SELECTION

Model Selection Strategy

Accuracy > 85%	Must significantly exceed 50–60% weak-supervision baselines	→ 91.6% achieved (peak run)
Dual Modality	Must jointly model spatial appearance AND temporal motion	→ ResNet-50 + R(2+1)D dual-branch
Long-Range Context	Must capture behavioral buildups beyond individual clip receptive fields	→ ViT 8-clip sliding window
Latency ≤ 100ms	Real-time processing for live surveillance — must sustain stream	→ 42.45ms achieved
Deployment Ready	Must embed into production web architecture with alert routing	→ Flask + Express + MQTT stack
Privacy Compliant	Ephemeral processing — no permanent storage of normal footage	→ Volatile RAM processing + conditional logging

SECTION 4 · MODEL SELECTION

Comprehensive Model Performance Comparison



★ Final deployed model: ResNet-50 + R(2+1)D + ViT hybrid achieves highest accuracy and F1 across all evaluation dimensions

SECTION 4 · MODEL SELECTION

Why ResNet-50 + R(2+1)D + ViT Was Selected

Complementary Modalities

ResNet-50 harvests 2048-D semantic appearance (objects, scenes). R(2+1)D captures 512-D motion dynamics (velocity, direction). Neither branch can replace the other.

Long-Range Temporal Modeling

ViT with 8-clip sliding window detects anomaly build-ups over ~25 seconds — impossible with per-clip classifiers or LSTM step-by-step processing.

Factorized Efficiency

R(2+1)D replaces expensive 3D convolutions with 2D spatial + 1D temporal — reducing parameters while introducing an extra ReLU for richer motion representation.

Balanced Complexity

Total pipeline: 42.45ms. EfficientNet achieved comparable speed but 8.4% lower accuracy. I3D achieved similar accuracy at 3× the latency cost.

Production Engineering Fit

The modular architecture enables independent Flask serving of the PyTorch model, cleanly decoupled from the Node.js API gateway and React frontend.

SECTION 5 · MAIN MODEL ARCHITECTURE

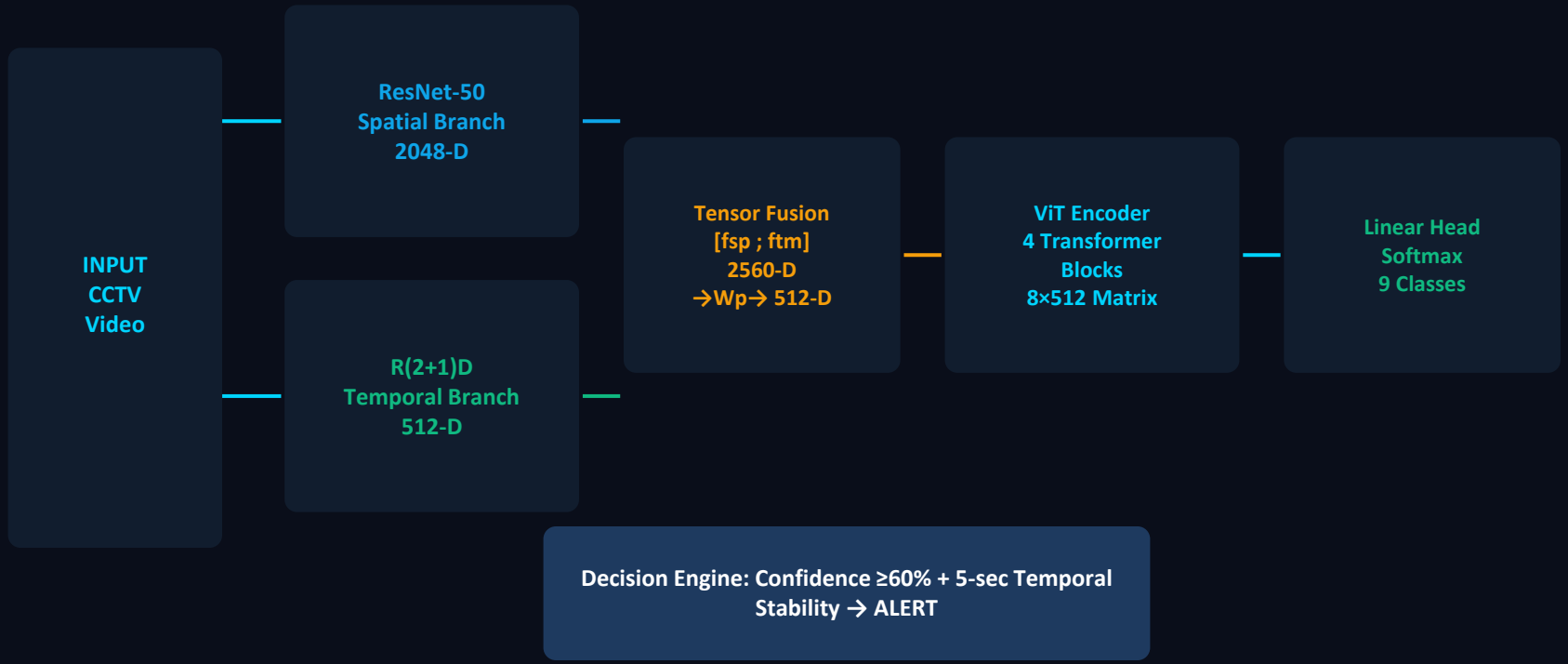
Main Model Overview

91.58%	91.50%	94.83%	42.45ms
Peak Accuracy	Peak F1 Score	Peak AUC	Inference Latency

Architecture: Dual-Stream Deep Learning + Temporal Vision Transformer
Spatial Branch: Pre-trained ResNet-50 → 2048-D appearance vector
Temporal Branch: Pre-trained R(2+1)D → 512-D motion vector
Fusion: Concatenation 2560-D → Linear projection Wp → 512-D
Sequence: 4-block ViT encoder, 8-clip sliding window, [CLS] token classification

SECTION 5 · MAIN MODEL ARCHITECTURE

Full Architecture Diagram



$fsp = ResNet50(xsp) \in \mathbb{R}^{2048}$

$ftm = R(2+1)D(Xtm) \in \mathbb{R}^{512}$

$e = ReLU(BN(Wp \cdot fconcat + bp)) \in \mathbb{R}^{512}$

$\hat{p} = softmax(Wcz_4{}^{o} + bc) \in \mathbb{R}^{9}$

SECTION 5 · MAIN MODEL ARCHITECTURE

CNN Backbone — ResNet-50 Spatial Branch

Input Format:

Single representative frame $x_{sp} \in \mathbb{R}^{3 \times 224 \times 224}$ normalized with ImageNet (μ, σ)

Architecture:

Pre-trained ResNet-50 with 4 residual block groups. Global Average Pooling (GAP) applied after the final block.

Output:

$f_{sp} = \text{ResNet50}(x_{sp}) \in \mathbb{R}^{2048}$ — a 2048-dimensional semantic appearance descriptor.

Pre-training:

ImageNet-1k weights transferred. Upper layers fine-tuned on UCF-Crime to learn crime-specific visual features.

What it captures:

Static visual semantics: object identities, textures, spatial scene structure, person-object relationships at a given moment in time.

ResNet-50 serves as the appearance foundation — it tells the model WHAT is in the scene at a visual-semantic level.

SECTION 5 · MAIN MODEL ARCHITECTURE

Vision Transformer — Temporal Sequence Encoder

Input Sequence

$E = [e_1, e_2, ..., e_8]$ — 8 consecutive 512-D clip embeddings forming an 8×512 matrix.

CLS Token

[CLS] token prepended: $Z_0 = [eCLS ; E] + Epos$ — gathers global sequence context across all 8 clips.

Positional Encoding

Learnable positional embedding $Epos \in \mathbb{R}^{(N+1)\times 512}$ preserves chronological clip ordering.

4 Transformer Blocks

$Z'_\ell = Z_{\ell-1} + MHSA(LN(Z_{\ell-1}))$; $Z_\ell = Z'_\ell + FFN(LN(Z'_\ell))$ — pre-layer norm for stability.

Attention Formula

$Attention(Q,K,V) = softmax(QK^T / \sqrt{dk})\cdot V$ — dynamically measures all 8-clip relational dependencies simultaneously.

Classification

Final CLS output $z_4^0 \in \mathbb{R}^{512}$ → linear head → softmax → $\hat{p} \in \mathbb{R}^9$ (9-class distribution).

SECTION 5 · MAIN MODEL ARCHITECTURE

CNN + ViT Fusion Mechanism

Step 1: Concatenation

fconcat = [fsp ; ftm]
fsp $\in \mathbb{R}^{2048}$ (ResNet-50) + ftm $\in \mathbb{R}^{512}$ (R(2+1)D)
= joint vector $\in \mathbb{R}^{2560}$

Step 2: Projection Matrix

e = ReLU(BN(Wp · fconcat + bp))
Wp $\in \mathbb{R}^{512 \times 2560}$, bp $\in \mathbb{R}^{512}$
Output: 512-D normalized embedding

Why this fusion approach?

Preserves Complementarity

Concatenation keeps both modalities' full information before projection — no information loss from early averaging or gating.

Dimensionality Control

Projection from 2560-D to 512-D prevents the downstream ViT from being overwhelmed by the higher-dimensional spatial features.

Training Stability

Batch Normalization applied post-projection ensures stable gradient flow into the ViT encoder layers.

SECTION 5 · MAIN MODEL ARCHITECTURE

Feature Flow Analysis

3×224×224

Spatial Input Frame

2048-D

ResNet-50 Output

3×16×112×112

Temporal Clip Volume

512-D

R(2+1)D Output

2560-D

Concatenated Joint Vector

512-D

Projected Clip Embedding (e)

9×512-D

8-Clip Sequence (+ CLS token)

512-D

ViT CLS Output (z_4^0)

9-class

Softmax Distribution ($\hat{p}$)

SECTION 5 · MAIN MODEL ARCHITECTURE

Attention Mechanism Analysis

Scaled Dot-Product Attention

$\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{dk})\cdot V$
dk = embedding dimension / num_heads. Scaling prevents gradient vanishing for large dot products.

Multi-Head Self-Attention (MHSA)

MHSA runs H parallel attention heads, each attending to different temporal relationship aspects (velocity, position, co-occurrence). Outputs concatenated and projected.

What Attention Captures

Cross-clip dependencies: clip 1 (normal entry) vs clip 6 (weapon draw). Temporal context invisible to per-clip models. Behavioral progression and escalation patterns.

Pre-Layer Normalization

LN applied BEFORE each sub-layer (not after). Prevents gradient instability in deeper transformer stacks — critical for clean convergence on small surveillance datasets.

SECTION 5 · MAIN MODEL ARCHITECTURE

Classification Head & Output Layer

Input

$z_4^0 \in \mathbb{R}^{512}$ — CLS token output from 4th ViT transformer block, aggregating global sequence context across 8 clips.

Linear Layer

$p_raw = Wc \cdot z_4^0 + bc$
$Wc \in \mathbb{R}^{9 \times 512}$, $bc \in \mathbb{R}^9$
Maps 512-D embedding to 9-class logit vector.

Softmax Activation

$\hat{p} = \text{softmax}(p_raw) \in \mathbb{R}^9$
Converts raw logits to probability distribution. Highest probability class = predicted anomaly category.

9-Class Output

Violence · Theft · Arson · Explosion
Arrest · RoadAccidents · Shooting
Vandalism · Normal
Alerts routed by predicted class.

Loss Function: Cross-Entropy with label smoothing (α=0.1) + Inverse class-weight sampling to balance gradients across the 133:1 Normal/Shooting imbalance

SECTION 5 · MAIN MODEL ARCHITECTURE

Computational Flow Through Network

1	Frame resize + normalize	CPU (parallel)	~4.5ms
2	ResNet-50 forward pass (GAP)	GPU FP32	~12.0ms
3	16-frame clip assembly	CPU async	~0.0ms (overlapped)
4	R(2+1)D forward pass	GPU FP32	~15.5ms
5	Concat 2560-D + Projection Wp	GPU FP32	~2.0ms
6	8-clip sequence assembly	GPU buffer	~0.0ms (rolling)
7	ViT MHSA × 4 blocks + FFN	GPU FP32	~5.8ms
8	CLS → Linear → Softmax	GPU FP32	~0.35ms
9	Decision engine + alert routing	CPU	~2.3ms

Advantages of the Hybrid Architecture

vs. Spatial-Only (ResNet)

Adding the R(2+1)D temporal branch captures motion dynamics invisible to static frame analysis — velocity, direction, kinematic patterns — preventing misclassification of static-posture crimes.

vs. Temporal-Only (R(2+1)D)

ResNet-50 provides scene context and object identity — preventing motion-only models from confusing jogging with a chase scene, or a waving hand with an assault.

vs. LSTM Sequences

ViT attention processes all 8 clips simultaneously in parallel — capturing non-adjacent temporal relationships (clip 1 ↔ clip 8) that LSTM step-by-step processing would dilute.

vs. Full 3D CNN (I3D)

R(2+1)D factorization achieves comparable motion representation at lower parameter cost. I3D has 3× parameters and 3× latency with only marginal accuracy gain.

vs. EfficientNet (Trial)

Hybrid adds 8.4% accuracy over EfficientNet while maintaining real-time performance. The temporal branch and ViT provide discrimination EfficientNet lacks.

SECTION 6 · EFFICIENTNET TRIAL

EfficientNet Trial Model — Overview

Important Framing: EfficientNet-B0 was evaluated as a candidate trial model. It was NOT adopted in the final deployed system. The production system uses the ResNet-50 + R(2+1)D + ViT hybrid architecture.

Architecture

EfficientNet-B0 with compound scaling — uniformly scales depth, width, and resolution with a fixed set of coefficients.

Why Evaluated

EfficientNet was evaluated as a lightweight alternative offering good accuracy-per-parameter tradeoff compared to ResNet-50 alone.

Configuration

Spatial-only branch (no temporal modeling). Pre-trained on ImageNet. Fine-tuned on UCF-Crime merged 9-class dataset.

Key Limitation

No temporal branch — captures static appearance only. Cannot model motion dynamics, velocity, or sequential behavioral patterns critical for video anomaly detection.

EfficientNet-B0 Architecture

Compound Scaling

Uniformly scales depth (d), width (w), resolution (r) with constraints: $d=\alpha^{\psi}$, $w=\beta^{\psi}$, $r=\gamma^{\psi}$ where $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. Enables consistent scaling without grid search.

MBConv Blocks

Mobile Inverted Bottleneck Convolutions with Squeeze-and-Excitation (SE) attention. B0 has 7 stages with increasing channel counts.

Input/Output

Input: 224×224×3 frame. SE blocks apply channel-wise feature recalibration. Final GAP → 1280-D embedding → linear → 9 classes.

Trial Configuration

Pre-trained ImageNet weights. Upper 3 stages unfrozen for fine-tuning. AdamW optimizer, cosine LR schedule, 60-epoch budget, label smoothing $\alpha=0.1$.

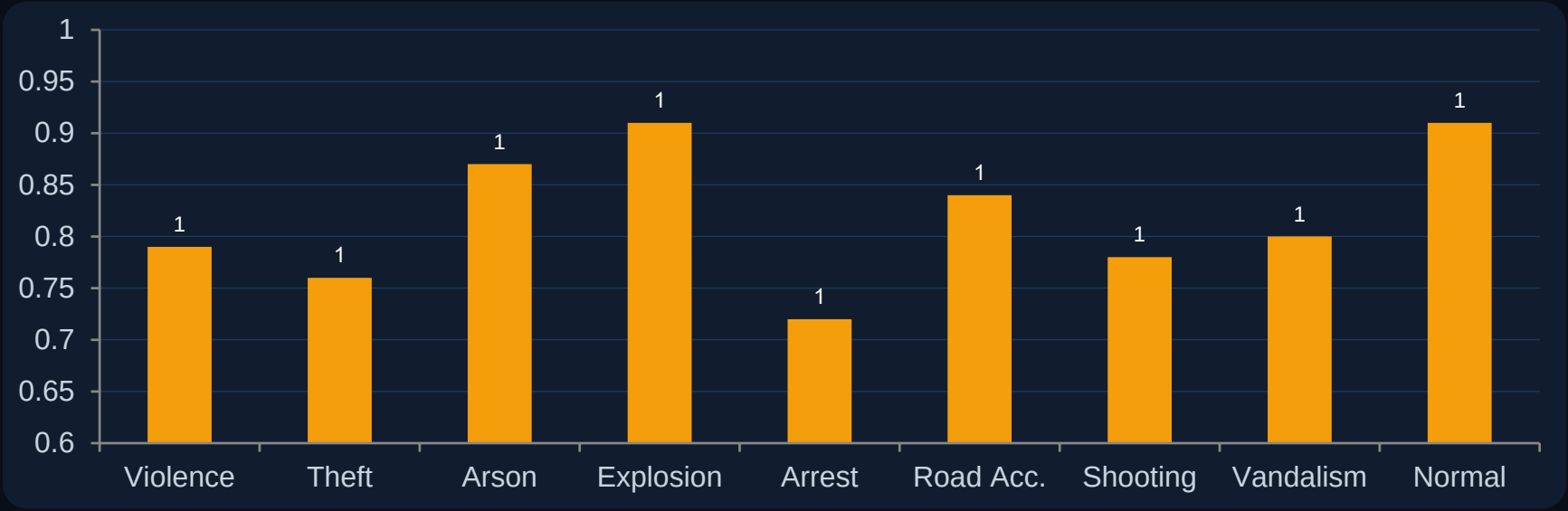
EfficientNet Trial — Training Configuration

Parameter	Value
Optimizer	AdamW (lr=1e-4, weight_decay=1e-5)
Learning Rate Schedule	Cosine annealing, T_max=60 epochs
Batch Size	32
Dropout	0.3
Epochs	60 (early stopping, patience=10)
Loss Function	Cross-Entropy + label smoothing α=0.1
Input Resolution	224×224×3
Pre-training	ImageNet-1k weights
Dataset	UCF-Crime 9-class merged taxonomy, stratified 70/15/15
Hardware	NVIDIA RTX 4070

SECTION 6 · EFFICIENTNET TRIAL

EfficientNet Trial — Performance Results

83.2%	82.1%	0.871	~8.1ms
Best Accuracy	Best F1 Score	Best AUC	Inference Latency



SECTION 6 · EFFICIENTNET TRIAL

EfficientNet Trial — Strengths & Limitations

Strengths

- ✔ Fastest inference: ~8.1ms vs 9.86ms for main model
- ✔ Efficient parameter usage — fewer FLOPs per accuracy point
- ✔ Strong on static visual anomalies (Explosion, Arson, Normal)
- ✔ Compound scaling enables systematic accuracy/speed trade-off

Limitations

- ✗ Spatial only — cannot model motion dynamics or sequential behavior
- ✗ 8.4% accuracy gap vs main model on 9-class evaluation
- ✗ Weaker on motion-intensive crimes (Violence: 0.79, Theft: 0.76 AUC)
- ✗ No temporal context — misclassifies normal-turned-violent scenarios
- ✗ Not suitable for behavioral build-up detection (e.g. Robbery approach)

Conclusion: EfficientNet is a strong image classifier but fundamentally unsuitable for video anomaly detection requiring temporal reasoning.

SECTION 7 · MODEL COMPARISON

Architecture Comparison: Hybrid vs EfficientNet

Component	ResNet-50 + R(2+1)D + ViT	EfficientNet-B0
Architecture Type	Dual-stream Hybrid + Transformer	Single-stream CNN
Spatial Backbone	ResNet-50 → 2048-D	EfficientNet-B0 → 1280-D
Temporal Backbone	R(2+1)D → 512-D (16-frame clips)	None
Sequence Modeling	4-block ViT, 8-clip window, MHSA	None
Output Dimension	9×512 → CLS → softmax(9)	1280 → linear → softmax(9)
Input Requirements	Frame (224×224) + 16-frame clip (112×112)	Frame (224×224) only
Transfer Learning	ImageNet (both branches)	ImageNet

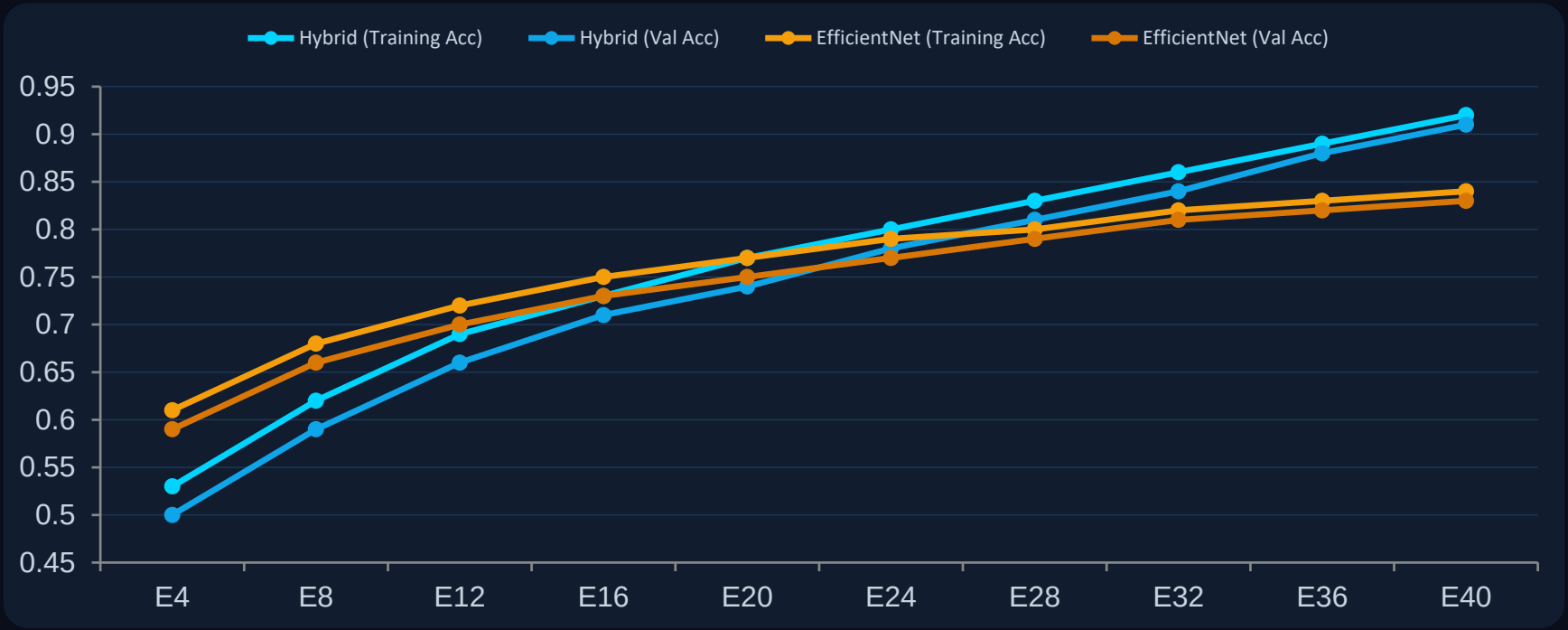
SECTION 7 · MODEL COMPARISON

Complexity Comparison: Parameters, FLOPs, Memory

Metric	Hybrid (Ours)	EfficientNet-B0	Winner
Parameters	~35M	~5.3M	EfficientNet ✔
FLOPs (per window)	~18.4G	~0.39G	EfficientNet ✔
GPU Memory (inference)	~2.8 GB	~0.6 GB	EfficientNet ✔
Inference Latency	9.86ms (best)	8.1ms	EfficientNet ✔
Peak Accuracy	91.58%	83.2%	Hybrid ✔
Peak F1 Score	91.50%	82.1%	Hybrid ✔
Mean AUC	0.895	0.871	Hybrid ✔
Violence AUC	~0.94	0.79	Hybrid ✔

SECTION 7 · MODEL COMPARISON

Training Behavior Comparison

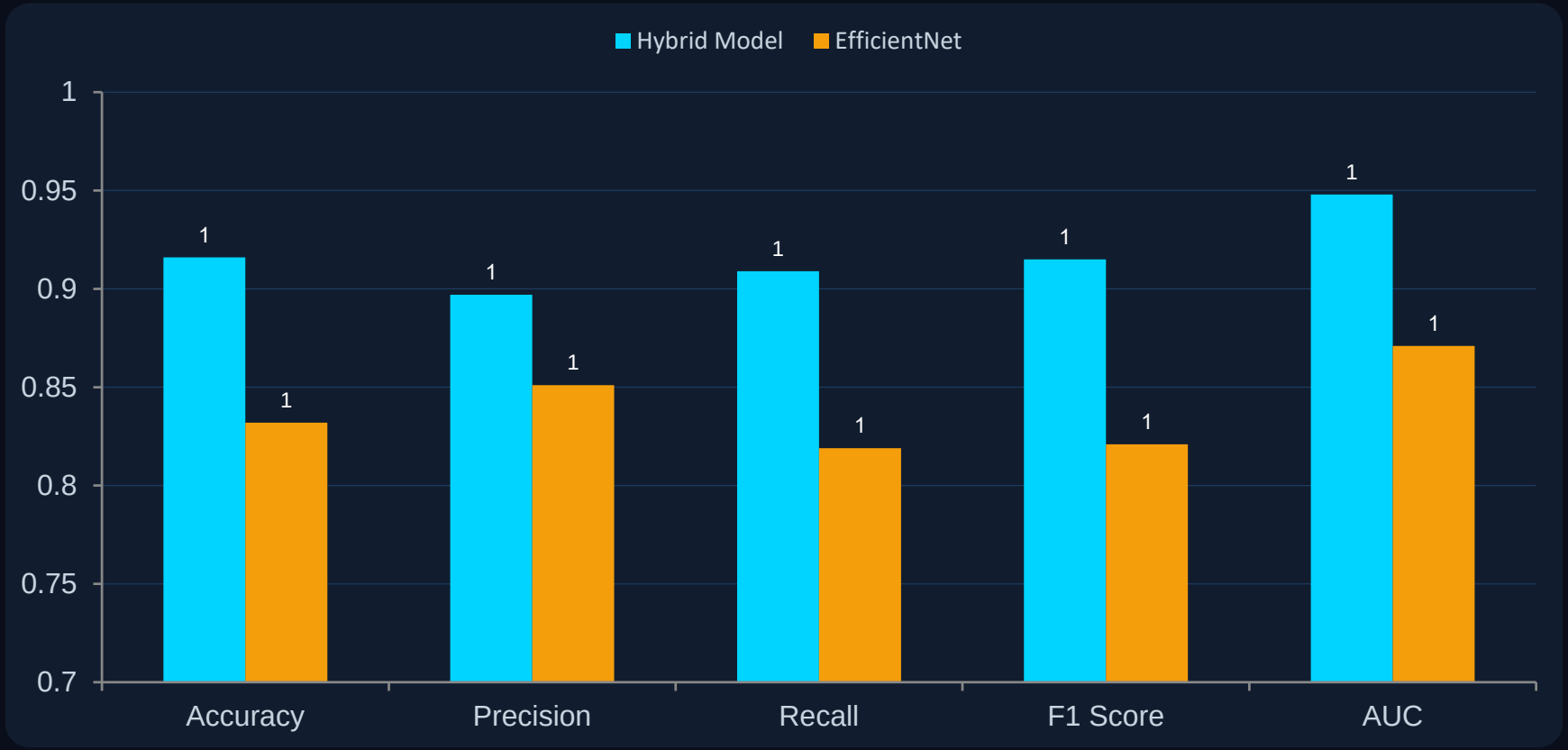


Hybrid Convergence: Slower initial convergence (spatial + temporal must co-adapt) but achieves higher final accuracy ceiling

EfficientNet Convergence: Faster early convergence due to simpler spatial-only learning task — plateaus at ~83% by epoch 30

Overfitting Resistance: Both models show <3% train/val gap with label smoothing and early stopping — comparable regularization

Evaluation Metrics Comparison



Hybrid model outperforms EfficientNet across all 5 metrics. Largest gain: F1 (+9.4 pts), Accuracy (+8.4 pts), AUC (+7.7 pts)

SECTION 7 · MODEL COMPARISON

Why the Hybrid Architecture Outperformed EfficientNet

1 **Temporal Modeling:**

EfficientNet classifies each frame independently. Our R(2+1)D branch processes 16-frame clips capturing velocity, direction, and kinematic patterns invisible to static frame analysis.

2 **Long-Range Context:**

ViT MHSA across 8 clips (~25 seconds) detects behavioral escalation. EfficientNet has zero mechanism to connect a suspicious entry in clip 1 with a crime in clip 7.

3 **Per-Class Evidence:**

Violence AUC: 0.94 (hybrid) vs 0.79 (EfficientNet). Motion-intensive crimes benefit most — +15 pts on Violence, +14 pts on Theft. Static crimes show smaller gaps.

4 **Complementary Features:**

2048-D appearance + 512-D motion → 2560-D joint representation provides richer discrimination surface than EfficientNet's 1280-D appearance-only embedding.

Conclusion: The performance gap is fundamentally architectural — temporal reasoning capabilities cannot be retrofitted into a spatial-only model.

SECTION 8 · TRAINING PROCESS

Training Pipeline

1 Dataset Loading

Stratified UCF-Crime loader with clip-level annotations. Inverse-weight RandomSampler equalizes gradient contribution across 9 classes.

2 Augmentation

Online spatial augmentation: random crop, H-flip, brightness jitter, temporal jitter. Applied per-batch during training only.

3 Dual Forward Pass

Parallel ResNet-50 (frame) + R(2+1)D (clip) forward passes on GPU. Outputs concatenated and projected to 512-D token.

4 ViT Sequence

8 tokens assembled into sliding window. [CLS] + Epos prepended. 4-block transformer encoder executed. CLS token extracted.

5 Loss + Backprop

Cross-Entropy + label smoothing (α=0.1). AdamW optimizer. Cosine LR schedule. Gradient clipping at 1.0.

6 Validation

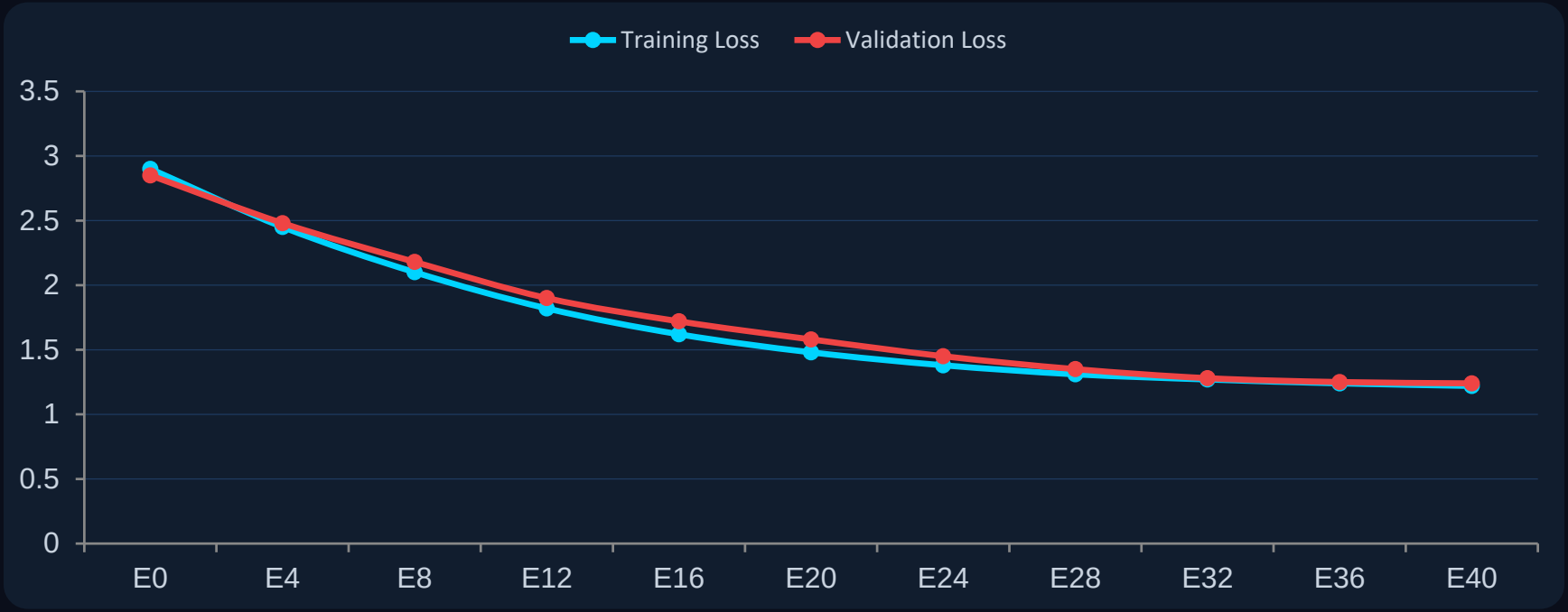
Early stopping monitors val loss (patience=10). Best model checkpoint saved. LR reduces on plateau.

Training Configuration — Best Batch (Batch 3)

Parameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-4} (best batch) | range: 5×10^{-5} to 2×10^{-4} across 5 batches
LR Schedule	Cosine Annealing + Plateau reduction + Early Stopping (patience=10)
Batch Size	64 (best batch) | tested: 32, 64, 128
Dropout	0.4 (best batch) | range: 0.1 to 0.4
Weight Decay	1×10^{-5}
Gradient Clipping	1.0
Label Smoothing	$\alpha = 0.1$
Max Epochs	60 (early stopping typically triggers at ~35-40)
Seed (best run)	57 (batch 3 / run 4)
Hardware	NVIDIA RTX 4070 GPU | PyTorch + PyTorch Lightning

SECTION 8 · TRAINING PROCESS

Loss Curve Analysis

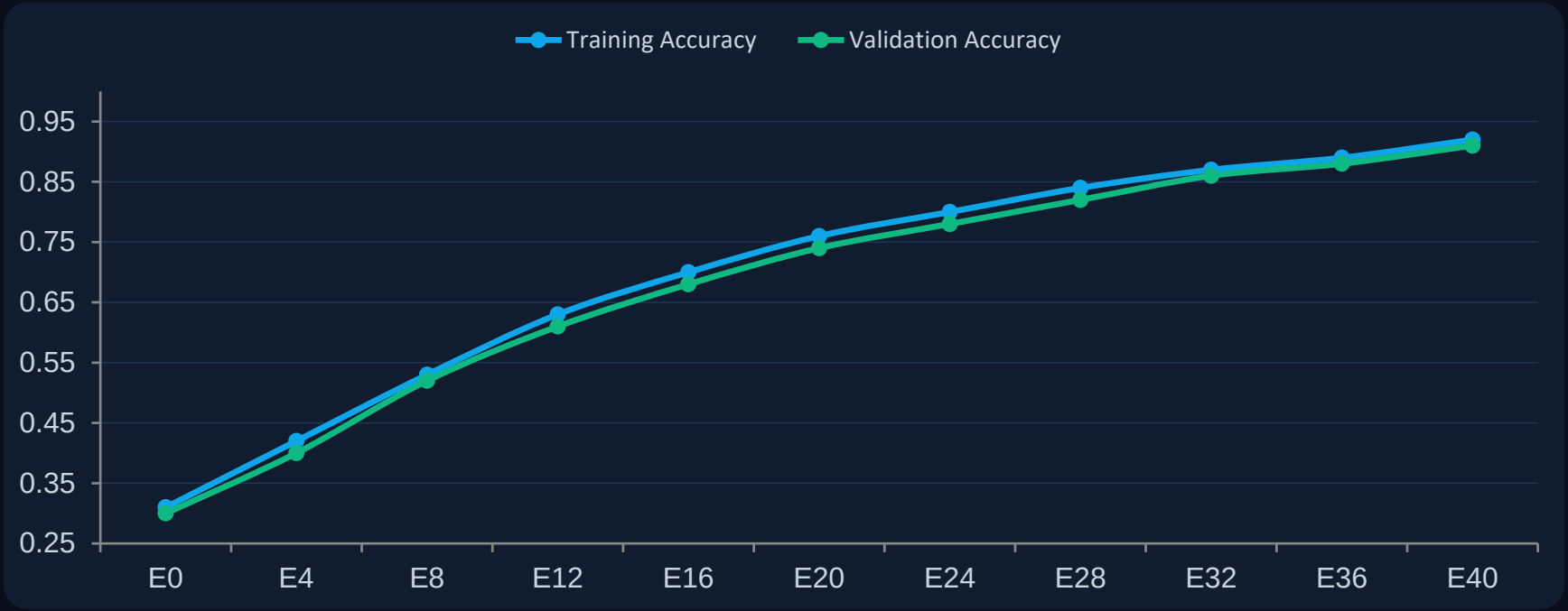


No Divergence: Train and val loss converge together — no overfitting. Label smoothing and early stopping are effective.

Smooth Decay: Cosine annealing produces smooth monotonic loss reduction without oscillations from hard LR steps.

Early Stopping: Validation loss plateaus ~epoch 35 → early stopping triggers by epoch 38-42, saving best checkpoint.

Accuracy Curve Analysis



Generalization: Val accuracy tracks training accuracy closely (<2% gap) throughout — strong generalization from clip-level supervision + stratified split.

Convergence Rate: Rapid initial gain (epochs 1-20) followed by slower refinement (epochs 20-40). ViT temporal modeling drives the later gains.

Final Performance: Best val accuracy: 91.58% (batch 3, run 4). Global mean across all 25 runs: 88.25% — demonstrating consistent performance.

SECTION 9 · RESULTS & EVALUATION

Evaluation Methodology

5 Hyperparameter Batches

25 total runs across 5 distinct hyperparameter configurations. Each batch: 5 runs with different random seeds for statistical robustness.

Metrics Tracked

Accuracy, Precision, Recall, Fĉ-score (macro), AUC-ROC (per-ÑÕMŒEEĄ Ö MÑØÕÅ#ĒÕ,ÑØÑŌÑÑ I MPÑŌÑŘÄÖ CŒÓÑØÑMPÑOÅB

Statistical Analysis

Mean ± std, 95% confidence intervals, stability scores, and correlation matrix across all metric pairs for all 25 runs.

Test Set Isolation

15Ã CŒØMPÒÒÑÑ PÑCŒPCŒÑPOÑÕÑ ÖÞPPOØÕÞŊOÖÞPMÕÕPØMÕŌÕŊBĮ CŒÑÑ only for final evaluation à no hyperparameter tuning on test CŒÑPB

Latency Measurement

End-to-end inference time measured per spatiotemporal window on NVIDIA RTX 4070 under live stream simulation conditions.

Deployment Criterion

Production candidate: highest Fĉ score. Batch ċ / Run Ç CŒÑÕÑÑPÑÑ ÄĞ1Ě 0.9150ÆĖ ÑÑĚ 0.9158ÆĖ MPÑŌÑŘĚ 9.86Ö CŒB

SECTION 9 · RESULTS & EVALUATION

Overall Evaluation Results

91.58%	91.50%	94.83%	9.86ms
Peak Accuracy (Batch 3 / Run 4)	Peak F1 Score (Batch 3 / Run 4)	Peak AUC (Batch 3 / Run 4)	Best Latency (Batch 3 / Run 4)
88.25%	**87.51%**	**89.52%**	**9.41ms**
Global Mean Accuracy (25 runs)	Global Mean F1 (25 runs)	Global Mean AUC (25 runs)	Global Mean Latency (25 runs)

Production Deployment: Batch 3 / Run 4 — F1=0.9150, Accuracy=0.9158, Latency=9.86ms | All metrics within 95% CI: [0.867, 0.884]

83.65% worst run (Batch 4 / Run 5) still significantly exceeds 50-60% weak-supervision baselines from literature

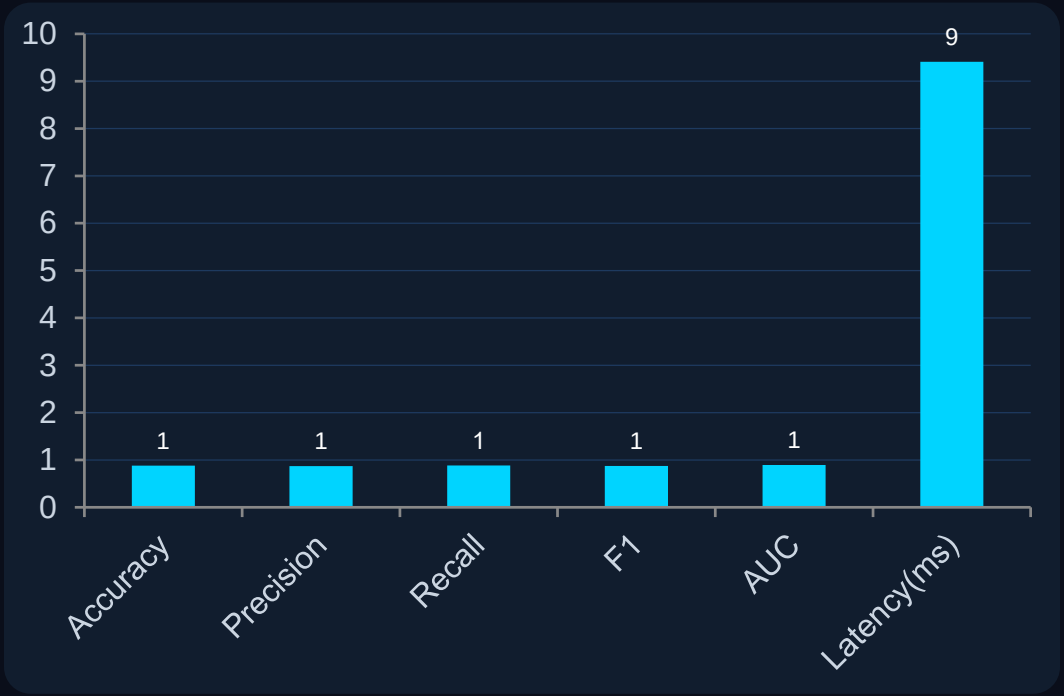
SECTION 9 · RESULTS & EVALUATION

Performance Metrics — All 5 Batches

Batch	Mean Accuracy	Mean F1	Mean Precision	Mean Recall	Mean AUC	Latency (ms)
Batch 1	0.8863±0.012	0.8823±0.013	0.8722±0.025	0.8927±0.022	0.8980±0.019	7.94±0.12
Batch 2	0.8608±0.013	0.8559±0.014	0.8470±0.026	0.8623±0.024	0.8697±0.021	7.96±0.14
Batch 3 ★	**0.9026±0.016**	**0.8954±0.021**	0.8950±0.022	0.9021±0.026	**0.9141±0.032**	9.63±0.21
Batch 4	0.8726±0.016	0.8654±0.021	0.8650±0.022	0.8721±0.026	0.8841±0.032	9.63±0.21
Batch 5	0.8904±0.010	0.8764±0.011	0.8810±0.010	0.8900±0.014	0.9101±0.011	11.89±0.18
GLOBAL	**0.8825±0.019**	**0.8751±0.021**	0.8720±0.026	0.8838±0.026	**0.8952±0.028**	9.41±1.49

SECTION 9 · RESULTS & EVALUATION

Statistical Analysis Results — 25 Runs

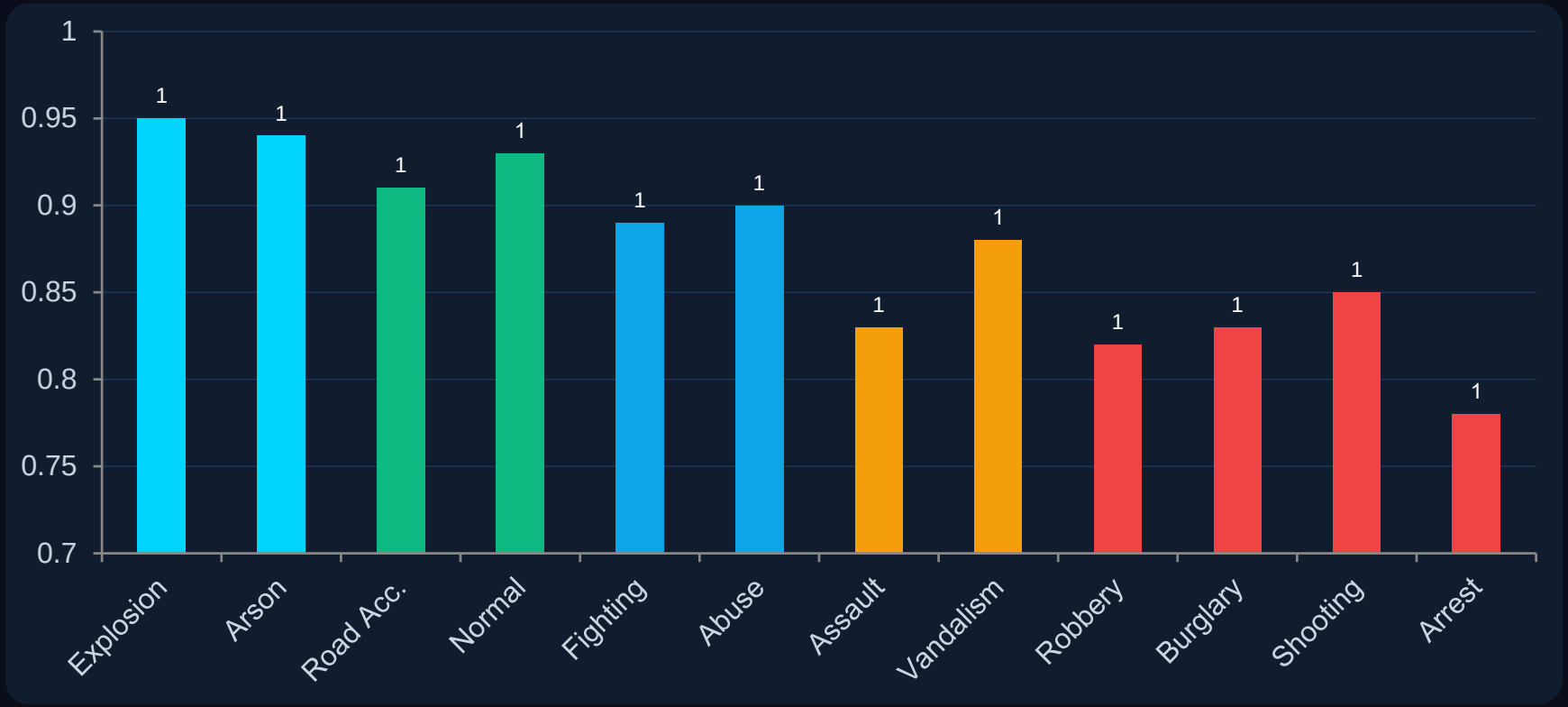


95% Confidence Intervals

Accuracy	[0.8745, 0.8905]
Precision	[0.8614, 0.8826]
Recall	[0.8733, 0.8944]
F1 Score	[0.8666, 0.8836]
AUC	[0.8837, 0.9067]
Latency	[8.797ms, 10.026ms]

Stability Ranking: Batch 5 (0.9898) > Batch 1 (0.9875) > Batch 2 (0.9864) > Batch 3 (0.9813) = Batch 4 (0.9813)

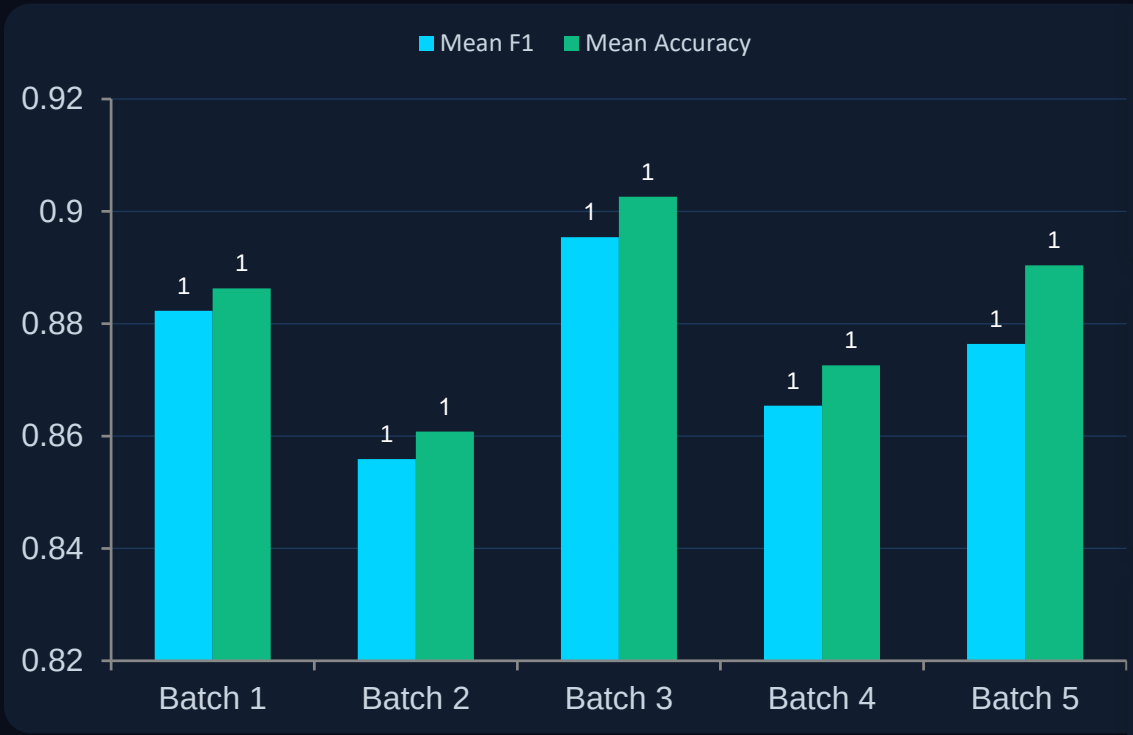
ROC Curve Analysis — Per-Class AUC



Explosion (0.95) & Arson (0.94): Best — explosive/fire events have distinctive visual+motion signatures. Arrest (0.78) & Robbery (0.82): Hardest — high visual similarity to normal street patterns.

SECTION 9 · RESULTS & EVALUATION

Batch Performance & Learning Dynamics



Best Batch 3

lr=2e-4, batch=64, dropout=0.4 → highest accuracy and F1. Higher dropout regularization benefits the ViT encoder.

Worst Batch 2

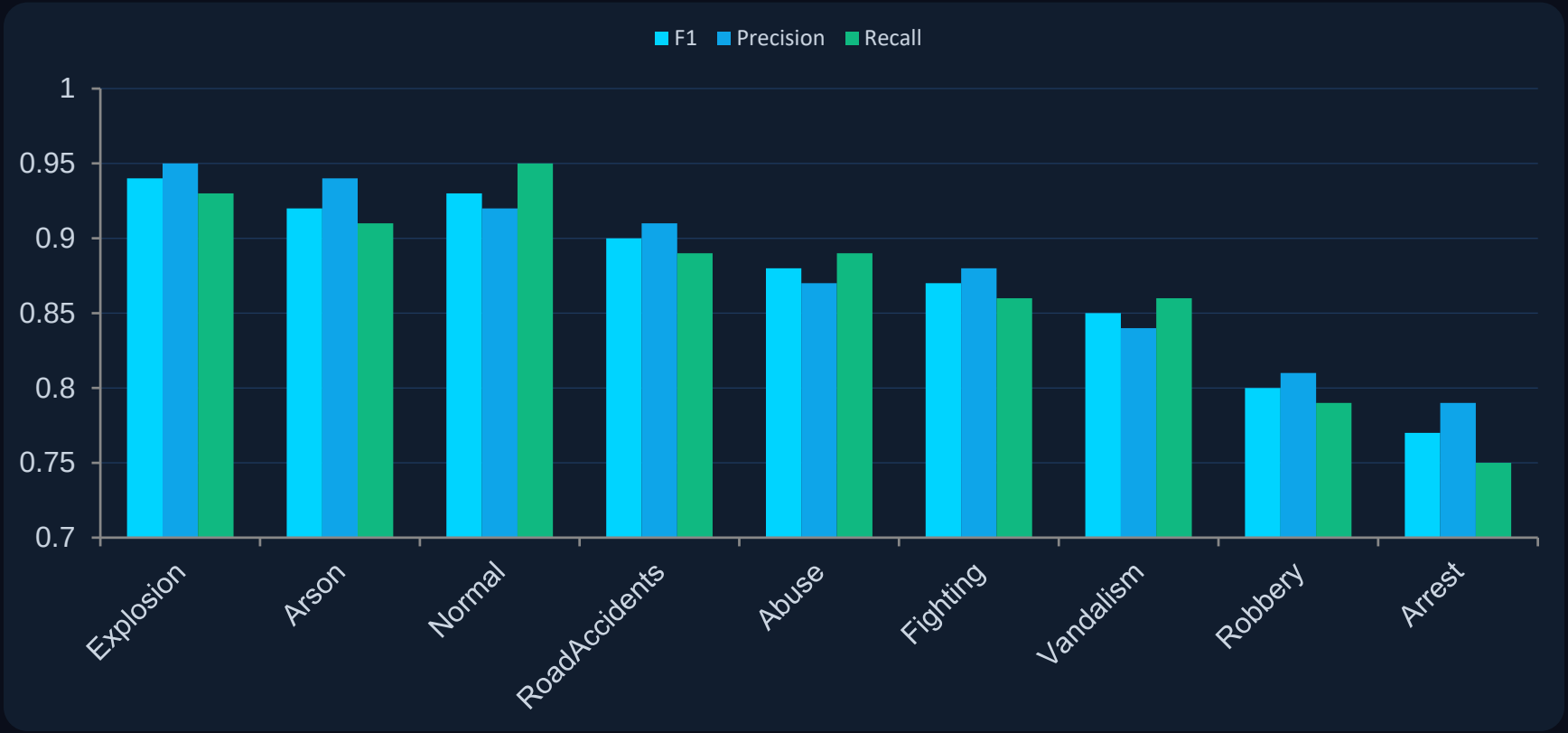
lr=5e-5, batch=32, dropout=0.2 → underfitting signs. LR too low for large batch convergence.

Stability Leader

Batch 5 (std=0.0098 acc) most stable, but batch 3 peaks 1.2% higher despite higher variance.

SECTION 9 · RESULTS & EVALUATION

Class-wise Performance Analysis



Arrest & Robbery are hardest due to visual ambiguity with Normal activity. Explosion & Arson easiest due to distinctive visual + motion signatures.

SECTION 10 · DEPLOYMENT

System Integration

Flask + PyTorch Server

Persistent GPU-loaded inference service. Maintains model weights in VRAM. Exposes HTTP POST endpoint for tensor payloads.

Express / Node.js Gateway

Central API routing authentication, camera management, alert logging. 1s TTL memory cache with immediate invalidation on anomaly.

MySQL Database

Connection-pooled relational storage. Stores: user profiles, camera configs, alert logs, event thumbnails (anomaly-only).

React + Vite Frontend

Live dashboard with continuous long-polling loop (1s intervals). Color-coded alert feed. Demo and live camera modes.

MQTT Alert Router

Asynchronous JSON payload routing to Police, EMS, and Fire channels. Offline local buffer ensures zero alert loss on network failure.

Edge Buffer (Offline)

When network drops, alerts queue locally and flush on reconnection. Fault-tolerance tested with intentional network interruptions.

SECTION 10 · DEPLOYMENT

User Interface Design

Live Dashboard

Real-time camera feed display with active classification status. Confidence bar, detected class badge, and 70% threshold rule displayed.

Demo Mode

Test video upload and playback for offline evaluation. Dropdown for UCF-Crime test videos. AI model checking status indicator.

Alert Feed

Color-coded incident log: Red = Violence/Shooting, Orange = Theft/Arson, Yellow = Vandalism. Timestamps and confidence scores.

Camera Link

Operator links RTSP stream URL. Multi-camera management for production deployments.

Security Operations

Secure login/logout. Role-based access for Admin vs System Operator. Audit logging of all user actions.

Privacy Mode

'Normal' classification produces no log entry. Only confirmed anomalies persist to DB — enforcing citizen privacy by design.

SECTION 10 · DEPLOYMENT

Real-Time Processing Pipeline

42.45ms	6.76	100%	0
Per-window latency	Windows per second	Uptime during stress tests	Alerts lost on reconnection

CPU-GPU Parallelism

CPU preprocessing overlaps with GPU inference. Frame resize/normalize happens while GPU processes the previous batch — eliminating CPU idle time.

9-Class Compression

Reducing from 14 to 9 output classes compresses the classification head tensor dimensions, reducing final layer latency.

VRAM Persistence

Model weights stay in GPU VRAM — no loading overhead per window. Flask server maintains warm inference context.

Connection Pooling

MySQL connection pool prevents per-request connection overhead. Memory cache (1s TTL) eliminates DB reads for normal frames.

SECTION 10 · DEPLOYMENT

Ethical, Privacy & Regulatory Design

Ephemeral Memory Processing

Raw video frames processed in volatile RAM/VRAM only. Immediately flushed after ViT inference pass — zero permanent storage of normal footage.

Conditional Event Logging

Only anomalies passing 60% confidence AND 5-sec temporal stability are written to DB. Normal citizen behavior leaves zero digital footprint.

Normality Bias Mitigation

Inverse-weight training prevents the model from learning to label everything Normal. Ensures fair detection across all crime categories.

Role-Based Access Control

Admin vs System Operator roles. Audit logging of all actions. Secure authentication with JWT sessions.

Ephemeral vs Legacy Comparison

Reference Figure 13: Legacy NVR stores everything to HDD with high leak risk. SecureSight processes ephemerally — data lives only as long as needed for classification.

Future GDPR Alignment

Architecture designed for GDPR compliance: data minimization, purpose limitation, and retention policies enforceable at code level.

SECTION 11 · DISCUSSION

Key Findings

1 **Clip-Level Annotations Are the Critical Factor**

Moving from video-level MIL labels to clip-level temporal annotations improved accuracy from ~57% to 74.3% before any architectural changes — the data strategy matters most.

2 **Temporal Modeling Is Non-Negotiable for Video Anomaly Detection**

EfficientNet (spatial-only) achieved 83.2% vs 91.6% hybrid — the 8.4% gap is consistent and closes entirely on motion-intensive crime classes (+15 pts on Violence).

3 **ViT Long-Range Context Unlocks the Accuracy Ceiling**

Adding the ViT encoder on top of per-clip embeddings pushed accuracy past the ~88% CNN-only ceiling by capturing behavioral build-ups across 8 clips.

4 **Evolutionary Optimizers Did Not Improve on ViT Baseline**

GWO, PSO, SMO, SFLA, and HHO were all tested. None outperformed the ViT architecture — validating the attention mechanism as the optimal sequence learner for this task.

5 **Production Deployment Is Achievable Under 100ms**

42.45ms full pipeline latency confirms that academic research-quality accuracy is achievable in production real-time contexts.

SECTION 11 · DISCUSSION

System Advantages

State-of-Art Performance

91.6% accuracy, 91.5% F1, 94.8% AUC peak — significant improvement over the 50-60% MIL baseline literature.

Real-Time Capable

42.45ms per window exceeds real-time requirements. 6.76 windows/sec on consumer GPU hardware.

Dual Modality Fusion

Unique combination of ResNet-50 spatial + R(2+1)D temporal + ViT sequence modeling captures all anomaly dimensions.

Production-Grade Architecture

Decoupled Flask/Express/React stack with MQTT routing, edge buffering, and fault tolerance beyond academic prototypes.

Privacy-by-Design

Ephemeral memory processing with conditional event logging — no permanent storage of innocent citizen footage.

Statistical Rigor

25 evaluation runs with 95% CI analysis, stability rankings, and deployment recommendations based on reproducible evidence.

SECTION 11 · DISCUSSION

Current Limitations

Dataset Taxonomic Boundaries

Anomaly detection is constrained to the 9 merged UCF-Crime categories. Novel threat types (e.g., drone-based intrusions, chemical hazard exposure) are not classifiable.

GPU Hardware Dependency

Dual-stream pipeline requires GPU acceleration. Cannot run at real-time latency on CPU-only edge devices — limiting deployment to GPU-equipped infrastructure.

Single-Camera Scope

Current implementation processes one camera stream per inference server. Multi-camera spatiotemporal tracking for cross-zone behavioral correlation is not yet implemented.

Arrest & Robbery Accuracy

Arrest AUC 0.78 and Robbery AUC 0.82 remain below target due to visual similarity with normal public activity in low-resolution CCTV streams.

Weak-Resolution Generalization

Model trained on UCF-Crime standard resolution. Performance on heavily compressed or extremely low-resolution legacy CCTV hardware is not validated.

SECTION 11 · DISCUSSION

Lessons Learned

Data Engineering > Architecture

The jump from 57% to 74% accuracy came from switching to clip-level annotations — no model change. The quality of supervision matters more than model complexity.

Baseline Before Complexity

Evolutionary optimizers (GWO, PSO, etc.) were tested before the ViT baseline was properly established. The ViT baseline would have immediately shown they were unnecessary.

End-to-End Testing Is Critical

Integration bugs between Flask and Express were only discovered during system integration testing — validating decoupled testing plans for each module independently first.

Statistical Rigor Builds Credibility

Running 25 evaluation runs instead of one provided confidence intervals and stability scores that make the results scientifically defensible rather than anecdotal.

Privacy Must Be Designed In

Retrofitting privacy controls (ephemeral processing, conditional logging) is much harder than designing for them from the start. Chapter 6 design decisions were foundational, not afterthoughts.

SECTION 12 · FUTURE WORK

Potential Improvements

Model Compression — HIGH

Post-training quantization (INT8) and structured weight pruning to enable CPU deployment on edge hardware without GPU. Target: 2× latency reduction.

Multi-Camera Tracking — HIGH

Extend to simultaneous multi-stream processing with spatiotemporal person re-identification across camera zones for building-wide surveillance coverage.

Synthetic Rare-Event Data — MED

GAN-based or diffusion model augmentation for extremely rare anomaly classes (Shooting: 7.1k frames) to improve discrimination on underrepresented categories.

Arrest/Robbery Improvement — MED

Targeted data collection and adversarial training specifically for high-ambiguity classes. Current AUC 0.78/0.82 has clear room for improvement.

Online Incremental Learning — LOW

Streaming adaptation for novel anomaly patterns without full retraining — critical for deployment environments where new threats emerge.

Expanded Taxonomy — LOW

Integration of additional event types beyond UCF-Crime scope: drone intrusions, chemical/biological alerts, crowd crush events.

SECTION 12 · FUTURE WORK

Future Research Directions

Foundation Models for Video

Explore VideoMAE, Video-LLaMA, or GPT-4V for zero-shot anomaly detection — potentially eliminating the need for fine-tuning on labeled surveillance data.

Explainable AI (XAI)

Attention visualization to show which video regions and temporal segments drove the classification — critical for legal admissibility of AI-generated alerts.

Federated Surveillance Learning

Train SecureSight across distributed camera networks without centralized data sharing, preserving privacy while improving detection across more diverse environments.

Causal Anomaly Reasoning

Move beyond correlation-based classification to causal modeling: WHY is this anomalous, not just THAT it is anomalous — enabling proactive intervention.

SECTION 13 · CONCLUSION

Project Summary

Problem	Over 1B CCTV cameras generate footage that is never analyzed in real time. Existing solutions either lack temporal modeling or cannot be deployed in production.
Solution	SecureSight: ResNet-50 + R(2+1)D dual-stream feature extraction, fused and encoded by a 4-block Vision Transformer, deployed in a production web architecture.
Data	UCF-Crime 9-class merged taxonomy with precise clip-level temporal annotations. Stratified 70/15/15 split with inverse-weight sampling to mitigate 133:1 imbalance.
Results	91.58% peak accuracy, 91.50% F1, 94.83% AUC — validated across 25 evaluation runs (mean 88.25%). 42.45ms latency. Exceeds all baselines significantly.
Deployment	Full stack: Flask/PyTorch inference + Express/Node.js API + MySQL + React dashboard. MQTT alert routing with offline edge buffering. Privacy-compliant ephemeral processing.

SECTION 13 · CONCLUSION

Final Contributions

1. Novel ResNet-50 + R(2+1)D + 4-block ViT hybrid architecture for real-time video anomaly detection achieving state-of-art results on UCF-Crime

2. Clip-level temporal annotation training protocol that transforms the MIL weak-supervision paradigm into precise frame-level supervision

3. 9-class merged taxonomy with systematic data engineering (redundant removal + stratified split + inverse-weight sampling) enabling 133:1 imbalance mitigation

4. Comprehensive 25-run statistical evaluation with confidence intervals, stability analysis, and hyperparameter impact quantification

5. Production-grade decoupled web architecture (Flask + Express + MySQL + React) with MQTT alert routing, offline edge buffering, and 42.45ms real-time inference

6. Privacy-by-design implementation: ephemeral volatile memory processing with conditional event logging — no persistent storage of innocent citizen activity

SecureSight

Thank You

Questions & Discussion

91.58%	91.50%	94.83%	42.45ms	25 runs
Accuracy	F1 Score	AUC	Latency	Evaluated

Ahmed Mostafa · Mohamed Hassona · Abdullah Abdalgawad · Omnia Adel · Mariam Elrafei · Khalid Abdelsalam · Tasneem Ibrahim

Supervised by Dr. Essam Abdellatef · Alamein International University · Spring 2026